

Image Processing (IMAJIS) – 2025-2026

2025/10/07 - Jean-Christophe Taveau - 1h00 - Documents allowed

0 1 7 6 4

2 2 1 1 3

1 7 4 2 5

3 5 2 1 6

Fig. 1: Image

3 6 9 10

0 1 8 6

2 2 1 1

1 15 14 12

Fig. 2: Image

0 0 0 0

① 1 0 0

0 1 1 1

0 0 0 ①

Fig. 3: Image

Q 1.1. What is the image size of Fig. 1?

Q 1.2. What is the minimal number of bits required for encoding the image of Fig. 1

(1-bit, 2-bit, 3-bit, ..., 8-bit, 9-bit, 10-bit, ..., 16-bit)? Justify your answer.

Q 1.3. What is the value of the pixel of coordinates (3,2) in image of Fig. 1?

Q 1.4. Transfer Function

1. Draw the default transfer function of Fig. 1.
2. If we add a value of 10 to all the pixel values of Fig. 1, draw the resulting transfer function? What is modified? the brightness or the contrast?
3. What kind of arithmetic operation on all the pixel values will modify the other parameter (brightness or contrast)?

Q 1.5. Extract all the one-bit channels from the image of Fig. 1.

This image may be considered as a color-like image composed of N one-bit channels where N is the number of bits required to encode this image. A one-bit channel is encoded as 1 bit. Calculate the N channels.

Q 1.6. Write the formula to normalize the image of Fig. 1 between 0 and 64?

1. Write the formula and give the calculations for the image of Fig. 1.
2. Draw the histogram.

Q 1.7. Convert the image of Fig. 2 into hexadecimal notation.

Q 1.8. Apply filters to the pixel of coordinates (1,2) in image of Fig. 1.

Give the intermediate calculations for all the filters.

1. A 3x3 square mean filter
2. A 3x3 median filter
3. A cross median filter

Q 1.10. Image padding in filtration

If we apply a 3x3 Gaussian filter to the image of Fig. 1 **without padding** the image, what will be the size of the output (filtered) image? Justify the answer.

Q 1.11. What is the result of the image segmentation by thresholding Fig. 1 at a value of 4 included?

Give the algorithm in pseudo-code and write the pixel values of the thresholded image.

Q 1.12. Count the number of objects from the image of Q. 1.11

Using the algorithm described in the lectures, draw the resulting image and corresponding data. Show two examples in the image of Q. 1.11 presenting the algorithm.

Q 1.13. What is the euclidean distance between two points P1 and P2 of XY-coordinates P1(0,1) and P2 (3,3)?

Give the formula and the intermediate calculations.

Q 1.14. What is the measured distance between the two circled points in Fig. 3?

The values 0=False and 1=True. Justify and compare to the result of Q 1.13.

Question 1.15. How is it possible to get the correct distance?

Justify and compare your results with Q 1.13 and 1.14.